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Paper: NDA 2024 (I) - Mathematics Authentic PYQ Full Test
Exam: NDA | **Type:** PYQ | **Time:** 150 Mins | **Max Marks:** 300.0 | **Negative Marking:** 0.83

Note: Defence Path is an independent educational portal. Authentic past questions compiled for candidate educational practice.

Q1. [Mathematics - Trigonometry] What is the value of sin(15°) + cos(15°)?

(A) 1 / √2
(B) √(3/2)
(C) √3 / 2
(D) √2

Correct Answer: *Option B*
Step-by-Step Explanation: *Step-by-step Solution:*
Let S = sin(15°) + cos(15°).
Squaring both sides: $S^2 = (\sin 15° + \cos 15°)^2 = \sin^2(15°) + \cos^2(15°) + 2 \sin(15°) \cos(15°)$
Since $\sin^2\theta + \cos^2\theta = 1$ and $2 \sin\theta \cos\theta = \sin(2\theta)$:
$S^2 = 1 + \sin(30°) = 1 + 1/2 = 3/2$.
Taking positive square root (as both sin 15° and cos 15° are positive in the first quadrant):
$S = \sqrt{(3/2)}$.
Therefore, Option B is correct.

Q2. [Mathematics - Quadratic Equations] If the roots of the equation $x^2 - px + q = 0$ differ by unity, then which one of the following relations is correct?

(A) $p^2 = 4q + 1$
(B) $p^2 = 4q - 1$
(C) $q^2 = 4p + 1$
(D) $p^2 + 4q = 1$

Correct Answer: *Option A*
Step-by-Step Explanation: *Step-by-step Solution:*
Let roots be α and β.
Given: $|\alpha - \beta| = 1$.
From Vieta's formulas for $x^2 - px + q = 0$:
Sum of roots: $\alpha + \beta = p$
Product of roots: $\alpha\beta = q$
We know that $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$.
Substituting values: $1^2 = p^2 - 4q$
=> $1 = p^2 - 4q$
=> $p^2 = 4q + 1$.
Hence Option A is correct.

Q3. [Mathematics - Calculus & Derivatives] What is the derivative of f(x) = ln(sec x + tan x) with respect to x?

(A) sec x
(B) tan x
(C) sec x tan x
(D) $\sec^2 x$

Correct Answer: *Option A*
Step-by-Step Explanation: *Step-by-step Solution:*
Using the chain rule of differentiation:
*d/dx [ln(u)] = (1/u) * du/dx.*
Here u = sec x + tan x.
$du/dx = \sec x \tan x + \sec^2 x = \sec x (\tan x + \sec x)$.
Therefore:
$d/dx [\ln(\sec x + \tan x)] = [\sec x (\sec x + \tan x)] / (\sec x + \tan x) = \sec x$.
Hence, the correct option is A.

Q4. [Mathematics - Matrices & Determinants] If A is a square matrix of order 3 such that |A| = 4, then what is the value of |adj(A)|?

(A) 4
(B) 16
(C) 64
(D) 12

Correct Answer: *Option B*
Step-by-Step Explanation: *Step-by-step Solution:*
For any square matrix A of order n:
|adj(A)| = |A|^(n - 1).
Here order n = 3 and determinant |A| = 4.
Therefore:
|adj(A)| = 4^(3 - 1) = 4² = 16.
Hence Option B is correct.

Q5. [Mathematics - Vectors & 3D Geometry] What is the angle between the two unit vectors a■ and b■ if |a■ + b■| = √3?

(A) 30° (π/6)
(B) 45° (π/4)
(C) 60° (π/3)
(D) 90° (π/2)

Correct Answer: *Option C*
Step-by-Step Explanation: *Step-by-step Solution:*
|a■ + b■|² = |a■|² + |b■|² + 2(a■ · b■)
Since a■ and b■ are unit vectors: |a■| = 1, |b■| = 1.
*Also, a■ · b■ = |a■||b■| cos θ = 1 * 1 * cos θ = cos θ.*
Given: |a■ + b■| = √3 => |a■ + b■|² = 3.
Therefore:
3 = 1 + 1 + 2 cos θ
3 = 2 + 2 cos θ
2 cos θ = 1 => cos θ = 1/2.
Since cos(60°) = 1/2, θ = 60° (or π/3 radians).
Hence Option C is correct.

Q6. [Mathematics - Probability] Two dice are thrown simultaneously. What is the probability that the sum of the numbers appearing on both dice is a prime number?

(A) 5 / 12
(B) 7 / 18
(C) 1 / 2
(D) 11 / 36

Correct Answer: *Option A*
Step-by-Step Explanation: *Step-by-step Solution:*
*Total sample space when 2 dice are rolled = 6 * 6 = 36 outcomes.*
Possible prime sums between 2 and 12 are: 2, 3, 5, 7, 11.
Outcomes giving sum = 2: (1,1) -> 1
Outcomes giving sum = 3: (1,2), (2,1) -> 2
Outcomes giving sum = 5: (1,4), (2,3), (3,2), (4,1) -> 4
Outcomes giving sum = 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) -> 6
Outcomes giving sum = 11: (5,6), (6,5) -> 2
Total favourable outcomes = 1 + 2 + 4 + 6 + 2 = 15.
Probability = 15 / 36 = 5 / 12.
Option A is correct.